

自适应游戏化学习系统中的学生画像本体 SPOnTo

Towards an Ontology for Representing a Student's Profile in Adaptive Gamified Learning System

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核心内容

Q：这篇论文做了什么？ A：提出了一个叫 SPOnTo 的学生画像本体，把学生的特征分成三大块——学业类、心理类、游戏化类——并用 Protégé 这个本体编辑器实现成一个 OWL 文件。它想同时解决两件事：一是自适应学习（按学生水平给内容），二是游戏化（按学生类型给激励元素）。

Q：它凭什么说自己比别人强？ A：作者的说法是前人只覆盖了学生特征的一部分。他们做了一张对照表，横轴是三类特征（学业、心理、游戏化），纵轴是九篇前人工作，结果是学业类特征几乎人人都用，心理类少数人用，游戏化类几乎没人用。SPOnTo 的卖点是三类都覆盖。

Q：这个本体的取值域是怎么定的？ A：这是全篇最值得一件事：每一个枚举取值域都来自一个有名字的、发表过的模型，没有一个是作者自己编的。学习风格用 Felder-Silverman 模型的四个维度（感知：感觉/直觉；处理：主动/反思；输入：视觉/言语；理解：序列/整体）；认知层级用布鲁姆分类法的六级（记住、理解、应用、分析、评价、创造）；人格用 FFM 五因素模型（外倾性、宜人性、尽责性、神经质、开放性）；玩家类型用 Hexad 六型（慈善家、破坏者、社交者、自由灵魂、成就者、玩家）。

Q：本体是怎么建起来的，有流程吗？ A：有，用的是 METHONTOLOGY 这套本体工程方法，作者把它简化成五步：确定范围与目标、知识获取、概念化、设计与实现、评估。关键在于「知识获取」被单列成一步，且排在「概念化」之前——先从领域专家和文献里把概念收上来，再去定类和属性，不能反过来。

Q：它怎么检验这个本体做得对不对？ A：用「基于准则的评估」，选了四条准则：清晰性、一致性、简洁性、正确性。每一条都给了在 SPOnTo 上真实改动过的例子。清晰性：`Student_information` 这个名字被专家否掉了，因为「学生」太泛，改成 `Learner_information` 才特指学习相关信息。一致性：`verbal` 和 `visual` 两个类原本同时挂在「学习风格」和「身体状况」下面，推理结果自相矛盾，于是从「身体状况」下删掉。简洁性：`Mental_capacity` 和 `Cognitive` 两个类职责重叠，合并成 `Cognitive_capacity`。正确性：专家反馈说必须加「教师反馈」和「家长反馈」两个概念，因为它们是识别学生画像并验证预测特征的关键。

Q：作者自己承认了什么局限？ A：本体还没在真实系统里跑过。结论一节说，下一步计划是把这个模型接进一个叫 class-quiz 的现有在线学习系统里去分析，才能验证有效性。也就是说这篇论文交付的是一个设计，不是一个被验证过的结果。

背景补充 / 点评

先补三个概念的出处，读这篇论文会反复碰到。

Protégé 是斯坦福大学做的免费开源本体编辑器，语义网领域事实上的标准工具，用它画类层级、定属性、跑推理机检查一致性。论文里说它「至今仍是使用最广泛的本体编辑环境」，这个判断在 2020 年是成立的。

METHONTOLOGY 是 1997 年马德里理工的 Fernández-López 等人提出的本体工程方法论，把本体开发当成软件工程来管，有明确的生命周期阶段。它之所以被选中，作者给的理由是「它是文献里最成熟的本体工程方法论之一，覆盖了本体开发生命周期里的大部分活动」。

布鲁姆分类法 (Bloom's taxonomy) 是 1956 年提出、2001 年修订的教育目标分类体系，把认知目标从低到高排成六级。它在教学设计里是通用语言，重要性在于它给「掌握到什么程度」提供了一个公认的刻度，而不是让每个系统自己定「初级、中级、高级」。

我的判断：这篇论文的本体内容对我们价值有限，但它的两个方法论细节价值很高。

先说价值有限的部分。它的画像属性清单是面向 K12 和游戏化激励场景的——玩家类型、审美偏好（颜色、字体）、身体状况、家长反馈——企业培训场景基本用不上。而且它有一个方法上的隐患：整个本体是从文献里拼出来的，九篇前人工作各贡献几个概念，作者自己也在概念表里如实标了每个概念来自哪一篇。拼装出来的本体覆盖面确实广，但没有一个统一的建模视角，所以才会出现 `Mental_capacity` 和 `Cognitive` 职责重叠这种问题——那是两篇不同论文的概念被同时收进来的后果。

再说价值高的两处。

第一处是概念表的第三列：来源。它的知识获取那一节有一张表，每一行是一个概念，三列分别是概念名、一句话描述、**来源文献**。所有取值域也都标了出处。这不是学术论文的引用规范问题，这是一条工程纪律：本体里的每一个维度和每一个取值，都要能回答「它从哪来」。一个说不清出处的维度，半年后没人敢删也没人敢用，本体就是这样腐烂的。

第二处是那四条评估准则，而且四条都配了真实改动过的例子，这让它从口号变成了可执行的检查项。尤其是一致性那个例子——`verbal`（言语型）既是学习风格又是身体状况，推理机跑出来矛盾——这说明他们真的开了推理机去查，不是人眼看一遍。

留白的地方：画像怎么被填上，论文几乎没写。它说数据来源是问卷、教务数据库、学习系统里的交互记录，但没有任何抽取机制的描述。画像随时间怎么更新、旧值怎么处理、冲突怎么消解，全部空白。所以这篇论文只能回答「画像长什么样」，不能回答「画像怎么活起来」。

对我自己的启示，有三条。

第一条最直接：**取值域必须来自一个有名字的、已经存在的模型或口径，不能自己编**。SPOnto 全篇没有一个自造的枚举——四个取值域全部来自 Felder-Silverman、布鲁姆、FFM、Hexad。这和我们项目里那条纪律（指标的定义只有四种合法来源：物理列取值域、数据里读出的阈值、页面在展示的口径、同一公式的分子分母）是同一件事，只是换了个领域。落到学员画像上，「掌握度」的分级不该我们拍脑袋定三档，要么用布鲁姆这类公认刻度，要么用考试配置里真实存在的及格线满分线。

第二条：**那四条评估准则可以直接做成画像本体的评审清单**。清晰性问的是这个名字有没有歧义（他们把 Student 改成 Learner 就是这一条命中）；一致性问的是同一个类挂在两个父类下会不会推出矛盾；简洁性问的是有没有两个类职责重叠（他们合并 Mental_capacity 和 Cognitive 就是这一条命中）；正确性问的是缺不缺关键概念。这四问的好处是它们互相不重叠，而且都能在 Protégé 或者我们自己的加载期校验里做成机械检查，不全靠人看。

第三条要警惕的：**这篇论文交付的是一个未经验证的设计，别把它的属性清单当结论抄**。它自己说了下一步才要接进真实系统。我们做画像本体如果照抄一张漂亮的类图，会重复它同一个风险——本体设计得很全，但没有一条属性被真实数据填过。对应到我们的做法上，应该是反过来：先看 Learning Lab 真实对话里学员实际说出了什么，再决定登记哪些维度。这也是 METHONTOLOGY 把「知识获取」排在「概念化」之前的原因。

文中金句

"The described ontology terms should communicate effectively the intended meaning, removing any aspects of subjectivity or ambiguity." 本体中描述的术语应当有效传达其意图含义，去掉一切主观性或歧义的成分。

"For instance, we had 'Student_information' as a concept... they agreed that the term 'Student' does not deliver its intended meaning because it's very general term." 举例来说，我们原本有一个概念叫 Student_information……专家们一致认为「Student」这个词没有传达出它本来要表达的意思，因为它是个非常笼统的词。

"Since we realized that categorizing cognitive is unnecessary and useless for our ontology, we have melted 'Mental_capacity' and 'Cognitive' classes into one class 'Cognitive_capacity'." 由于我们意识到对「认知」再做分类对本体既无必要也无用处，我们把 Mental_capacity 和 Cognitive 两个类融合成了一个类 Cognitive_capacity。

"Most gamified systems follow the 'one size fits all' approach when integrating elements of the game, without taking into account differences between learners." 大多数游戏化系统在引入游戏元素时采用的是「一套打天下」的做法，没有考虑学习者之间的差异。

一、引言

Online learning platforms, educational systems and continuing education programs are all based on a uniform approach that may be inappropriate to the profile of the learners, where a large number of students are educated without taking into account their preferences, their cognitive abilities, their learning style, their behaviors and their very particular personalities typical to each one of them.

在线学习平台、教育系统和继续教育项目都建立在一种一刀切的做法之上，而这种做法对学习者的画像可能并不适配：大量学生在接受教育时，他们的偏好、认知能力、学习风格、行为，以及各自非常独特的个性，都没有被考虑进去。

In fact, a new concept appeared called adaptive learning connected to artificial intelligence and cognitive science research, which provides learning appropriate to a learner's ability and takes into account his profile and his preferences. This concept has been introduced by many researchers as a solution for students' engagement but it is not as effective as anticipated. The lack of motivation is one of the main reasons for this issue.

事实上，出现了一个与人工智能和认知科学研究相关联的新概念，叫做自适应学习：它提供与学习者能力相匹配的学习，并把学习者的画像和偏好考虑在内。很多研究者把这个概念当作解决学生投入度问题的方案引入，但它的效果并不如预期。动机不足是造成这一问题的主要原因之一。

However, motivation is a decisive factor in students' learning that determines how much effort and commitment they have put into learning a particular topic, and the human being only carries out any activity taking into account the reward obtained once this activity is successfully completed. In this regard, many researchs have shown a link between motivation and gamification. This latter is the use of game metaphors, game elements and ideas in a different context than games to increase motivation and engagement, as well as to influence user behavior.

然而，动机是学生学习中的决定性因素，它决定了学生在学习某个特定主题时投入多少努力与承诺；而人只有在考虑到活动成功完成后所获得的回报时，才会去执行某项活动。在这方面，很多研究表明动机与游戏化之间存在联系。所谓游戏化，是把游戏隐喻、游戏元素和游戏思路用在游戏之外的语境中，以提升动机与投入度，并影响用户行为。

Several researchs approved that the integration of gamification has great potential to engage students and facilitate learning but it is complicated to implement it properly and it is not a simple process of adding game elements. Most gamified systems follow the "one size fits all" approach when integrating elements of the game, without taking into account differences between learners. So, to target these problems, we have proposed "SPOnto": an ontology of

representation of student profile where we combined the two concepts "adaptive learning" and "gamification".

若干研究确认，引入游戏化在吸引学生投入和促进学习方面有很大潜力，但要正确实施却很复杂，它不是简单加几个游戏元素的过程。大多数游戏化系统在引入游戏元素时采用的是「一套打天下」的做法，没有考虑学习者之间的差异。因此，针对这些问题，我们提出了 SPOnTo：一个表示学生画像的本体，在其中我们把「自适应学习」和「游戏化」这两个概念结合起来。

The comparative study of Hamim et al shows that the ontology approaches are the most recently used to model the profile of student. This latter was applied for the purpose of modeling the student profile, as being a technique that can give as complete a representation as possible of the student profile. Ontology has been commonly used as a useful knowledge engineering technique to reduce ambiguity and help with information sharing.

Hamim 等人的对比研究表明，本体方法是近期用于给学生画像建模的最主流方法。本体之所以被用来给学生画像建模，是因为它是一种能对学生画像给出尽可能完整表示的技术。本体一直被广泛用作一种有用的知识工程技术，用来减少歧义、帮助信息共享。

二、相关工作

The use of ontologies in student profiling has gained considerable attention in recent years since it is suitable to tackle interoperability and information sharing challenges. Several ontological models have been proposed in the literature to overcome these issues.

近年来，本体在学生画像中的应用受到相当多关注，因为它适合应对互操作性与信息共享方面的挑战。文献中已经提出了若干本体模型来解决这些问题。

Munassar and Ali proposed a Framework for adaptive e-learning (PALO) which is based on semantic web technology and offers students personal materials. They modeled learning objects and the learner's profiles according to different learning styles and knowledge levels by integrating the OWL ontology and the SWRL rules. The work presented by Abyaa et al based on modeling the learner's knowledge using ontologies and rule reasoning. They take into consideration in their adult learner's knowledge model the different knowledge types and categories, learner's prior knowledge, previously learned but forgotten knowledge, misconception and errors.

Munassar 和 Ali 提出了一个自适应在线学习框架（PALO），它基于语义网技术，为学生提供个性化材料。他们通过集成 OWL 本体和 SWRL 规则，按照不同的学习风格和知识水平对学习对象和学习者画像建模。Abyaa 等人的工作基于用本体和规则推理给学习者的知识建模。他们在成人学习者知识模型中考虑了不同的知识类型与类别、学习者的先验知识、曾经学过但已遗忘的知识、以及误解与错误。

Ameen et al proposed student profile ontology to personalize the content learning, based on the academic information. Sarwar et al concentrated on different attributes to profile the learner

such as learning style, knowledge, age, locale/origin, professional experience and qualification. They proposed a semantic e-learning framework not only for profiles learners through ontology but also for categorizing them based on their profiles for recommending the suitable learning content.

Ameen 等人提出了一个学生画像本体，基于教务信息来个性化学习内容。Sarwar 等人聚焦于用不同属性来刻画学习者，比如学习风格、知识、年龄、地域/出身、职业经历和资历。他们提出的语义化在线学习框架不仅用本体刻画学习者，还基于画像对学习者的分类，以推荐合适的学习内容。

According of the description of existing studies related to student profile ontology, we were allowed to highlight the major limit that was identified is that the most studies didn't give a complete representation of student profile in all the aspects related to the student features. The majority has concentrated in one or some characteristic of the student and didn't cover all the different aspects in the same time. We can notice that the academic features are the most used. The majority of studies didn't focus enough on student motivation, engagement and how can use gamification features to more attract them. The source of data varies from one study to another, and the majority of studies use questionnaires, academic databases, and the interactions of learners for the e-learning systems.

根据对现有学生画像本体相关研究的描述，我们能够指出被识别出的主要局限：大多数研究并没有在与学生特征相关的所有方面上给出完整的学生画像表示。多数研究集中在学生的某一个或某几个特征上，没有同时覆盖各个不同方面。可以注意到，学业类特征是被使用得最多的。多数研究对学生动机、投入度、以及如何用游戏化特征吸引学生这几件事关注不足。数据来源各研究之间差异很大，多数研究使用问卷、教务数据库，以及学习者在在线学习系统中的交互记录。

The added value of our work in comparison with the previous mentioned works is that we propose a student profile ontology that captures the full details and aspects related to the student features, to give as complete a representation as possible of the student profile, which containing both implicit and explicit information about him. We defined a taxonomy inspired by Hamim et al to represent the different characteristics and categorize them in three categories: academic, psychological and gamification features.

与前述工作相比，我们工作的增值之处在于：我们提出的学生画像本体捕捉了与学生特征相关的全部细节和方面，以给出尽可能完整的学生画像表示，其中同时包含关于学生的隐式信息和显式信息。我们定义了一套受 Hamim 等人启发的分类法来表示这些不同特征，并把它们归为三类：学业类特征、心理类特征、游戏化类特征。

三、SPOnto 本体的开发

Ontologies can be built from scratch or can be reused an existing ontologies. There are many methodologies for ontology development. Among them we were inspired by a METHONTOLOGY to develop our SPOnto ontology from scratch, simplifying the process of five main steps that

can be described as follow: (1) Scope and requirements definition, (2) Knowledge acquisition, (3) Conceptualization, (4) Design & Implementation, (5) Evaluation. We choose this methodology because it is listed as one of the most mature ontology engineering methodologies existing in literature.

本体可以从零构建，也可以复用已有本体。本体开发有很多方法论，其中我们受 METHONTOLOGY 启发，从零开发 SPOnTo 本体，并把流程简化为五个主要步骤：(1) 范围与需求定义；(2) 知识获取；(3) 概念化；(4) 设计与实现；(5) 评估。我们选择这套方法论，是因为它被列为文献中现存最成熟的本体工程方法论之一。

3.1 范围与目标的确定

The scope of our ontology is the learner's profiling. The main users of our ontology are schools and e-learning systems. The main objectives of our ontology are: Offering an efficient description of students in various aspects (behavior, knowledge, learning style etc.) in order to help decision-making in the different academic context and to be able to act in case of problems such as failure, drop out. Define the most adaptive learning resources depending on their profiles. Therefore, a profile model allows the detection of profile patterns which can be applied in similar cases which leads to facilitate the research, accelerate the decision and make it more efficient and make the system centered on the user.

我们本体的范围是学习者画像。本体的主要使用者是学校和在线学习系统。主要目标有两个：其一，在多个方面（行为、知识、学习风格等）对学生给出有效的描述，以便在不同的教学场景中辅助决策，并在出现挂科、辍学这类问题时能够采取行动。其二，根据画像确定最适配的学习资源。因此，一个画像模型使我们能够发现画像模式，这些模式可以应用到相似情形中，从而便于检索、加快决策、提升效率，并让系统以用户为中心。

3.2 知识获取

Once we have established the scope and the objectives of our ontology, we move to the second step of the pre-development stage which is "knowledge acquisition". This step comprises exploration, extraction and derivation of knowledge from the domain of interest. In our case learner's profiling, we have acquired the domain knowledge and key concepts and elements from domain experts and researchers working in this field. We used terminologies from prominent existing ontologies from literature.

一旦确定了本体的范围和目标，我们就进入开发前阶段的第二步，即「知识获取」。这一步包含从关注领域中探索、抽取和推导知识。在我们这个学习者画像的场景中，我们从领域专家和该领域的研究者那里获取了领域知识与关键概念和要素。我们采用了文献中已有的重要本体所使用的术语。

（原文表 2 列出从文献中抽取的学生画像概念，每个概念给出描述与来源文献。概念包括：个人信息，描述传记与人口统计学数据；人格，作为使个体行为在时间上保持一致的内部因素；成绩单，描述以机构为基础的学业成就汇总；动机，即做某事的意愿，可分为内在动机和外在动机；表现，描述学习者被测量到的表现，例

如成绩；胜任力，描述学习者在学习过程中获得的知识、技能与能力的集合；学习风格，描述学习者偏好的学习与获取知识的方式，也包括其身体上和社会性上的需要；认知，指与信息操作相关的各类心理过程，包括智力、胜任力、经验、能力；技能与兴趣，指使学生得以成功的才能与兴趣中心，包含创造力、人际、沟通、领导力、理解力（速度与逻辑）；知识，指通过经验或教育获得的、关于某一特定主题的信息，可分为一般的、理论的、实践的；资历，描述授予学习者的任何资格、证书、执照或学位；玩家类型，指在游戏化系统面前对用户进行分类并识别彼此差异；游戏元素，指把游戏中的任何类型的概念、机制等应用到游戏化中。）

3.3 概念化

Next phase, following the METHONTOLOGY process, is to perform the conceptualization of our ontology. This step takes as input the list of concept resulting from the knowledge acquisition. It includes defining the core concepts, a glossary of terms, a tree of concepts, and the relations between the concepts in the ontology. Based on our sources of knowledge, we defined the following core concepts: Profile, which represents all the basic information relating to student's background, goals, interest and preferences. Personal Information, which represents the biographic and demographic data. Academic Information, which represents all information related to the student learning, such as the information acquired through experience or education, thinking abilities and skills etc. Player Information, which represents all information related to the player such as player type and game elements preferences etc.

按照 METHONTOLOGY 流程，下一阶段是对本体做概念化。这一步以知识获取阶段产出的概念清单作为输入，内容包括定义核心概念、术语词表、概念树，以及本体中各概念之间的关系。基于我们的知识来源，我们定义了以下核心概念：Profile（画像），表示与学生背景、目标、兴趣和偏好相关的全部基础信息；Personal Information（个人信息），表示传记与人口统计学数据；Academic Information（学业信息），表示与学生学习相关的全部信息，例如通过经验或教育获得的信息、思维能力与技能等；Player Information（玩家信息），表示与玩家相关的全部信息，例如玩家类型和游戏元素偏好等。

The glossary of terms contains the definitions of all terms related to the domain (concepts, attributes, relations). Table 3 provides a list of some concepts used in our ontology which we specify, for every concept, the following information: the properties, the parent, and the relationships.

术语词表包含与该领域相关的全部术语（概念、属性、关系）的定义。原文表 3 给出了本体中部分概念的清单，对每个概念都写明三项信息：属性、父类、以及关系。举例来说，Learning style（学习风格）这一概念的父类是 Learner_information，它的关系有 hasLearning_Style、hasInput_LearningStyle、hasPerception_LearningStyle、hasProcessing_LearningStyle、hasUnderstanding_LearningStyle 五条，属性是 Learning_style_value；Cognitive（认知）的父类是 Learner_information，关系有 hasDisabilities、hasAbilities、hasCompetences、hasIntelligence，属性是 Cognitive_Type 与 Cognitive_Value；Interaction preference（交互偏好）的父类是 Player_information，关系有 hasAestheticPreferences 及其颜色、字体两个细分，属性是 Color_value 与 Font_value。

3.4 设计与实现

Our ontology is implemented with protégé 5.2 ontology editor, which is by far the most widely used ontology editing environment, and is saved as an OWL (Web Ontology Language) file. OWL is a common ontology language which defines and describes classes, subclasses and properties i.e., object properties, datatype properties and annotation properties.

我们的本体用 Protégé 5.2 本体编辑器实现——它至今仍是使用最广泛的本体编辑环境——并保存为 OWL (Web 本体语言) 文件。OWL 是一种通用的本体语言，用于定义和描述类、子类和属性，即对象属性、数据类型属性和标注属性。

Learner Information, which represents the academic information. Students' academic details are collected in "Learner_information" class which is in turn divided into several subclasses such as "Learning_style", "Mental_capacity", "Knowledge", "Skills", "Cognitive" and "Learning_goals". Among these subclasses we find: Learning style, which describes the learner's preferred method of learning and acquiring knowledge. There are several models of learning style, but we used in our work "Felder Silverman Learning Style Model" which contains a four-dimensional: Dimensions Perception (sensing / intuitive), Dimensions Processing (active / reflective), Input measurement (visual / verbal), and Understanding (sequential / global).

学习者信息表示的是学业信息。学生的学业细节被收纳在 Learner_information 类中，该类又分成若干子类，例如 Learning_style (学习风格)、Mental_capacity (心智能力)、Knowledge (知识)、Skills (技能)、Cognitive (认知) 和 Learning_goals (学习目标)。在这些子类中：学习风格描述学习者偏好的学习与获取知识的方式。学习风格有多个模型，但我们在工作中使用的是 Felder-Silverman 学习风格模型，它包含四个维度：感知维度 (感觉型/直觉型)、处理维度 (主动型/反思型)、输入方式 (视觉型/言语型)、理解方式 (序列型/整体型)。

Mental capacity, which represents a hierarchy of educational objectives such as cognitive, sensory and affective domains. For this criterion we used the bloom taxonomy, which helps the instructors to analyze the level of each student under six categories of cognitive domains such as remembering, understanding, applying, analyzing, evaluating and creating. This taxonomy not only helps student to evolve thinking abilities but also to identify the skills they are lacking with.

心智能力表示的是教育目标的层级体系，涵盖认知、感官和情感领域。对这一项准则我们使用布鲁姆分类法，它帮助教师在六类认知领域下分析每个学生的水平，这六类是记住、理解、应用、分析、评价、创造。这套分类法不仅帮助学生发展思维能力，也能识别出他们所缺失的技能。

Personal Information, which represents the biographic and demographic data. This includes data such as: name, age and address etc. All of these data are collected in "Personal_information" class which is in turn divided into several subclasses such as "Physical_Condition", "Personality" and "Demographic_data". For the personality criteria, we followed the FFM model, which categorizes personality traits in five dimensions: (1) Extraversion; (2) Agreeableness; (3) Conscientiousness; (4) Neuroticism; and (5) Openness.

个人信息表示传记与人口统计学数据，包括姓名、年龄、地址等。这些数据全部收纳在 Personal_information 类中，该类又分成若干子类，例如 Physical_Condition（身体状况）、Personality（人格）和 Demographic_data（人口统计学数据）。对人格这一项准则，我们遵循 FFM 模型，它把人格特质归为五个维度：(1) 外倾性；(2) 宜人性；(3) 尽责性；(4) 神经质；(5) 开放性。

Player Information, which represents all information related to the player such as player type, game elements preferences etc. These information are collected in "Player_information" class which is in turn divided into several sub classes such as "Interaction_preferences", "Motivation" and "Player_type". For the player type classification, we used a Hexad typology, which contains six types of players: (i) Philanthropists, (ii) disruptors, (iii) Socializers, (iv) Free Spirits, (v) Achievers and (vi) Players.

玩家信息表示与玩家相关的全部信息，例如玩家类型、游戏元素偏好等。这些信息收纳在 Player_information 类中，该类又分成若干子类，例如 Interaction_preferences（交互偏好）、Motivation（动机）和 Player_type（玩家类型）。对玩家类型的分类，我们使用 Hexad 类型学，它包含六种玩家类型：(i) 慈善家；(ii) 破坏者；(iii) 社交者；(iv) 自由灵魂；(v) 成就者；(vi) 玩家。

3.5 评估

In the literature, many approaches of ontology evaluation have been suggested to assess whether ontology accurately and properly represented such domain information. We have reviewed the most common evaluation approaches and we have chosen the criteria-based evaluation approach. We selected a list of criteria among those such as clarity, consistency, conciseness and correctness.

文献中已经提出了很多本体评估方法，用来判断本体是否准确恰当地表示了相应的领域信息。我们回顾了最常见的几种评估方法，选择了基于准则的评估方法。我们从中选取了一组准则：清晰性、一致性、简洁性、正确性。

Clarity: the described ontology terms "should communicate effectively the intended meaning" removing any aspects of subjectivity or ambiguity. The ontology should be documented with a natural language. In our context, the feedback of the interviewed domain experts has helped us to verify ontology clarity and more specifically to omit or replace any ambiguous term. For instance, we had "Student_information" as a concept which represents all information related to the student learning. During an interview with experts of the domain, they agreed that the term "Student" does not deliver its intended meaning because it's very general term which can represent any detail of student and should be replaced with the term "Learner" that is more significant and refers specifically to learning information of the student.

清晰性：本体中描述的术语「应当有效传达其意图含义」，去掉一切主观性或歧义的成分。本体应当有自然语言的文档。在我们的情境中，受访领域专家的反馈帮助我们检验了本体的清晰性，更具体地说，帮助我们删掉或替换了有歧义的术语。举例来说，我们原本有一个概念叫 Student_information，用来表示与学生学习相

关的全部信息。在与领域专家访谈的过程中，专家们一致认为「Student」这个词没有传达出它本来要表达的意思，因为它是个非常笼统的词，可以指代学生的任何细节；应该换成「Learner」这个更有意义、且专门指向学生学习信息的词。

Consistency, or coherence, requires the logical coherence of ontology concepts and elements and avoids contradiction or ambiguities. Any contradiction between explicit or inferred axioms and their definitions makes the ontology incoherent. As an example, the class "verbal" and the class "visual" were subclasses of both "Learning_style" and "Physical_condition". In one hand, these classes are defined as a learning style and in another hand; these are defined as a type of physical condition. However, the inferences were inconsistent. So, we removed these classes from "Physical_condition".

一致性（或称融贯性）要求本体的概念与元素在逻辑上融贯，避免矛盾或歧义。显式公理或推理得出的公理与它们的定义之间只要出现任何矛盾，本体就是不融贯的。举个例子，verbal（言语型）类和 visual（视觉型）类原本同时是 Learning_style（学习风格）和 Physical_condition（身体状况）两个类的子类。一方面这两个类被定义为一种学习风格，另一方面它们又被定义为一种身体状况类型。然而推理结果是不一致的。因此我们把这两个类从 Physical_condition 下移除了。

Conciseness, means that there should be no unnecessary concepts or explicit redundancies between definitions in the ontology. For instance, regarding SPOnTo ontology, it included 2 classes "Mental_capacity" and "Cognitive" as subclasses of the class "Learner_information". "Mental_capacity" represented educational objectives such as cognitive, sensory and affective domains, based on the six cognitive categories of bloom taxonomy and "Cognitive" described some of mental processes such as intelligence and competence etc. Since we realized that categorizing cognitive is unnecessary and useless for our ontology, we have melted "Mental_capacity" and "Cognitive" classes into one class "Cognitive_capacity".

简洁性意味着本体中不应存在不必要的概念，各定义之间也不应有显式的冗余。举例来说，SPOnTo 本体原本在 Learner_information 类下有两个子类 Mental_capacity 和 Cognitive。Mental_capacity 基于布鲁姆分类法的六个认知类别，表示认知、感官、情感领域这类教育目标；Cognitive 则描述智力、胜任力等一些心理过程。由于我们意识到对「认知」再做分类对本体既无必要也无用处，我们把这两个类融合成了一个类 Cognitive_capacity。

Correctness, means that the ontology represents the correct modeling of the real-world concepts. This criterion has been carefully considered in SPOnTo ontology development and validation. As an example, domain experts' feedback has led us to add the concepts "Teacher feedback" and "Parent feedback" since they constitute a key factor in student profile identification and validation of his predicted characteristics.

正确性意味着本体对现实世界概念的建模是正确的。这条准则在 SPOnTo 本体的开发与验证中被仔细考虑过。举例来说，领域专家的反馈使我们加入了「教师反馈」和「家长反馈」两个概念，因为它们构成了识别学生画像、以及验证对学生预测出的特征这两件事上的关键因素。

四、结论

In the learning system, the major problem facing students is that they cannot obtain pertinent information based on their requirements which in turn produces a feeling of boredom and reluctance in the students and decreases their sense of motivation and engagement. Among the solutions associated with content personalization in learning systems, dealing with the heterogeneity of student profile in different aspects is a complex task. But modeling an efficient student profile describes the best way a student prefers to learn and reflects his true needs which in turn would enhance the usage.

在学习系统中，学生面临的主要问题是他们无法根据自身需求获得切题的信息，这反过来使学生产生厌倦与抗拒，并削弱他们的动机感和投入感。在学习系统内容个性化的各种解决方案里，处理学生画像在不同方面上的异质性是一项复杂任务。但是，建好一个有效的学生画像能刻画出学生偏好的最佳学习方式、反映他真实的需要，而这反过来会提升使用效果。

The paper proposes "SPOnto": an ontology of representation of student profile in a learning system which connects two concepts "gamification" and "adaptive learning". Our ontology allows to build a global model of the student based on many important characteristics in order to help to predict their intentions and preferences and the decision-making to personalize the learning scenario. The resulting ontology was evaluated by virtue of a criteria-based approach to check its design and content. In future work, we intend to apply our model in an existing e-learning system, called "class-quiz", to analyze it in a real system and approve the efficiency of the student profile model on the basis of all these characteristics.

本文提出 SPOnto：一个在学习系统中表示学生画像的本体，它把「游戏化」和「自适应学习」两个概念连接起来。我们的本体使我们能基于许多重要特征构建学生的整体模型，以帮助预测学生的意图和偏好，并辅助个性化学习场景的决策。所得到的本体通过基于准则的方法进行了评估，以检查其设计与内容。在未来工作中，我们打算把这个模型应用到一个叫 class-quiz 的现有在线学习系统中，在真实系统里分析它，以证实基于这全部特征的学生画像模型的有效性。

参考链接

- [Protégé 本体编辑器](#) — 斯坦福大学的开源本体编辑器，论文用其 5.2 版本实现 SPOnto，可画类层级并跑推理机查一致性
- [METHONTOLOGY 方法论原文](#) — Fernández-López 等人 1997 年提出的本体工程方法论，论文简化成五步流程使用
- [Felder-Silverman 学习风格模型](#) — 论文学习风格取值域的出处，北卡州立大学 Felder 教授的官方资料页